HOTEL MANAGEMENT SYSTEM

Software Requirements Specification

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2 INTRODUCTION

2.1 PURPOSE

This Software Requirements Specification (SRS) will focus on Hotel Management System (HMS). This system will streamline hotel activities like booking a room, guest check in and check out, billing and generating reports. The aim of this SRS document will be to minimize manual tasks, improve the overall processes and hence enhance customer satisfaction of hotel guests. This document will help in guiding all the project stakeholders so that the system being designed meets with the requirements in perspective and within the scope of the hotel. It will support design, quality assurance as well as deployment to facilitate successful roll out.

2.2 SCOPE

This document focuses on hotel room booking, customer management, billing, inventory and reports. Note that such third-party integrations are out of scope (e.g. travel agencies, gateways to third parties e.g. payments platforms)

2.3 DEFINITIONS, ACRONYMS, AND ABBREVIATIONS

- **HMS:** Hotel Management System
- **OTA:** Online Travel Agencies
- **API:** Application Programming Interface
- **DB:** Database
- **POS:** Point of Sale
- **SSL:** Secure Socket Layers
- **TLS:** Transport Layer Security
- **JSON:** JavaScript Object Notation
- **XML:** Extensible Markup Language
- **UAT:** User Acceptance Testing

3 OVERALL DESCRIPTION

3.1 PRODUCT PERSPECTIVE

The Hotel Management System (HMS) is expected to operate independently by automating the main activities of the hotel such as making the reservations, checking in, checking out different guests, generating the bills, and preparing reports. It will, however, make use of internal hotel systems such as inventory and financial systems which will integrate well to offer all the important services in one application. The HMS will facilitate operational activities and ensure proper data movement between different processes or departments and there is room for further expansion of such systems incorporating 3rd party services such as OTA.

3.2 HIGH-LEVEL SUMMARY

Real-time room booking and reservation management
Guest check-in/check-out handling
Billing and payment processing
Room status updates
Reporting for management

3.3 USER CLASSES

User Class	Description	Access Requirements
Hotel Staff	Hotel employees handling bookings, check-ins, and billing	Need access to booking and check-in/out functionalities
Hotel Manager	Responsible for overseeing operations and analyzing reports	Access to reports, system monitoring, and analytics
Guests	End users making reservations and checking booking statuses	Make reservations, view their bookings online
IT Admin	Responsible for maintaining the system and handling security	System configuration, user management, backup control

4 OPERATING ENVIRONMENT

4.1 HARDWARE

Desktops and tablets used by staff, public kiosks for guest self-check-ins, and server infrastructure for data storage.

4.2 SOFTWARE

The HMS will run on a web-based application accessible through browsers (Google Chrome, Microsoft Edge, Mozilla Firefox, etc.) with a backend that integrates with hotel financial systems and databases (MySQL).

4.3 DEPENDENCIES

Stable internet connectivity for real-time bookings, POS systems for payments, and regular system backups to prevent data loss.

5 SPECIFIC REQUIREMENTS

Requirement ID	Description	Priority
FR-01	The system must allow online and in-person room reservations.	High
FR-02	Room availability must be updated in real time after each booking.	High
FR-03	System must generate bills for guests at checkout, including additional services	Medium
FR-04	The system keeps track of the amount of inventories used.	High
FR-05	Reports on occupancy and revenue must be available to the manager.	High
NFR-01	The system must handle 100 concurrent users without performance issues.	High
NFR-02	All sensitive guest data must be encrypted.	High
NFR-03	The system must support easy retrieval and updating of data from the database	High
NFR-04	Ensure 99.9% uptime with automatic backups.	High

5.1 FUNCTIONAL REQUIREMENTS

5.1.1 ROOM RESERVATION SYSTEM

Scenario: Guests can request and book rooms using the hotel's website/media contact of the hotel or by a hotel Receptionist.

Requirement: It is up to date on current room status.

Desired Outcome: It presents a range of rooms available for booking, chooses the type of room required and books it with the booking information.

Result: Room status is changed to reserved, and guest information is captured into the system.

5.1.2 CHECK-IN AND CHECK-OUT PROCESS

Scenario: The staff members of the hotels can check guests in and out, with the status of the room changing automatically through the system.

Requirement: The Guest should have made a booking.

Desired Outcome: The system detects the check-in of the guest, changes the room occupation status to occupied and renders checkout functions including billing.

Result: After checkout of the clients, the occupancy of the room is either changed to available or it will require cleaning.

5.1.3 BILLING AND PAYMENT HANDLING

Scenario: For example, as a front desk staff, at check out when a guest wants to settle their bill, I want to complete the guests bill with any additional costs consumed from a meal served in the room.

Desired Outcome: The system can give the total bill inclusive of all the additional charges made and produces a simplified bill for the customer and collects payments through cash and credit cards among others.

Result: Records for payments are done and invoices issued.

5.1.4 INVENTORY MANAGEMENT

Scenario: As a manager of the housekeeping department, he/she must see to it that the room supplies can be tracked and replenished whenever necessary.

Desired Outcome: The system keeps track of the number of rooms supplies used and gives a warning when the level goes low.

Result: Inventory levels are being updated in the system in real time.

5.1.5 REPORTING AND ANALYTICS

Scenario: A manager of the hotel needs to create performance-related reports on guest numbers and acquisitions, revenues coming from hotel services and feedback from the guests.

Desired Outcome: The system can pull up daily, weekly, and monthly reports showing the rate of room occupancy, levels of income generated, and other key performance indicators.

Result: Reports were developed in PDF format or any available formats for authorized users.

5.2 NON-FUNCTIONAL REQUIREMENTS

5.2.1 PERFORMANCE

The system should support up to 100 users connecting concurrently, the response to the request should be delivered within 2 seconds.

Must handle bookings and check in and out of over 500+ rooms.

5.2.2 SECURITY

Encryption of Data: All the most sensitive data collected from the guests like their details and even credit card information will be encrypted regardless of whether it is stored or in transit, for example using SSL or TLS.

5.2.3 USABILITY

The system must be user-friendly to the users for easy access to retrieval and updating of data from the database like the booking, check in / out and billing. The system should be able to operate effectively in a web-based environment on a computer and on a mobile phone allowing staff to access data easily for operations

5.2.4 RELIABILITY

The system availability should be at least 99.9 % that entails better performance of continuous operations in the system.

5.2.5 ASSUMPTIONS

Databases can be deployed through cloud-based infrastructure or the traditional based infrastructure which have a reasonable assurance of reliability and are managed by the IT department of the hotel.

There would be security measures put in relation to the database that will include encryption and automatic backups.

5.2.6 DEPENDENCIES

HMS will be depending on the database in storing and managing all sensitive data including guests' information, bookings, inventories, and billings.

System performance is reliant on the databases of the systems that guarantee concurrent query support to ease the hotel operations.

5.3 EXTERNAL INTERFACE REQUIREMENTS

5.3.1 USER INTERFACE (UI)

An intuitive web-based system will be implemented, which may be accessed from desktops and mobile devices. Moreover, it is necessary for the users of the UI including reservations, billing, and check-in to carry out executable functions.

Simple layout and easy navigation of the UI had to be ensured for the non-technical users such as hotel Staff and guests.

5.3.2 HARDWARE INTERFACES

The system should be compatible with the following hotel hardware components:

- **POS systems:** For payments collection.

- **Printers:** For issuing bills and receipts for the guests.
- **Public kiosks:** For self-service check-in.

5.3.3 SOFTWARE INTERFACES

This system shall interact with:

- **SQL Database:** To hold information on the hotel, its bookings, payments, and guests.
- **Financial Systems:** For relevant billing and payment data integration.

5.3.4 COMMUNICATION INTERFACES

The system will use standard protocols (e.g., HTTPS for secure web communication) to interact with external systems and services, ensuring secure and encrypted data transmission.

JSON or XML may be used for data exchange between the HMS and third-party services.

6 SYSTEM FEATURES

Feature	Description	Acceptance Criteria
Room Booking and Reservation	Guests can make online and offline reservations.	Booking confirmations sent instantly.
Check-in and Check-out	Hotel staff can manage guest check-ins and check-outs.	Only confirmed bookings can proceed to check-in.
Billing and Payment	Automatic bill calculation at check-out.	Final bill must include all room rates and extra services (e.g., dining).
Reporting	Generates reports for hotel managers showing occupancy, revenue, and inventory.	Reports must be available within 30 seconds of the request.

7 ACCEPTANCE CRITERIA

The requirements of both functional and non-functional types need to pass through UAT.
For system acceptance:

- All the functionality is tested and meets the functional requirements.
- The non-functional requirements (performance, security, etc.) pass/complete stress and security tests.
- The actual users (hotel staff, managers, etc.) approve the usability and performance during the UAT phase.

8 APPENDICES

8.1 GLOSSARY OF TERMS

- HMS (Hotel Management System): provide tailored solutions to keep your hotel business operating optimally.
- OTA (Online Travel Agencies): is an online or app-based marketplace where consumers can browse, book, and pay for hotel rooms, transportation, lodging, airfare, restaurants, and experiences.
- API (Application Programming Interface): is a collection of communication protocols and subroutines used by various programs to communicate between them.
- DB (Database): is an organized collection of data stored in a computer system and usually controlled by a database management system (DBMS).
- POS (Point of Sale): It is a computer-based cash register with software capable of tallying up orders, taking payments, monitoring inventory and buying trends, creating invoices, and collecting marketing data.
- SSL (Secure Socket Layers): is a security protocol that provides privacy, authentication, and integrity to Internet communications. SSL eventually evolved into Transport Layer Security (TLS).
- TLS (Transport Layer Security): is a security protocol that provides privacy and data integrity for Internet communications. Implementing TLS is a standard practice for building secure web apps.
- JSON (JavaScript Object Notation): is a standardized format for representing structured data.
- XML (Extensible Markup Language): is a software- and hardware-independent tool for storing and transporting data.
- UAT (User Acceptance Testing): is the final testing stage in software development before production. It’s used to get feedback from users who test the software and its user interface (UI).

9 RELATED DOCUMENTS

Software Requirements Specification (SRS): Its presentation specifies system features and requirements for HMS.
System Design Document: Describes features of system architecture and design related decisions.
User Manual: Written for hotel personnel concerning the operation of the Hotel Management system.

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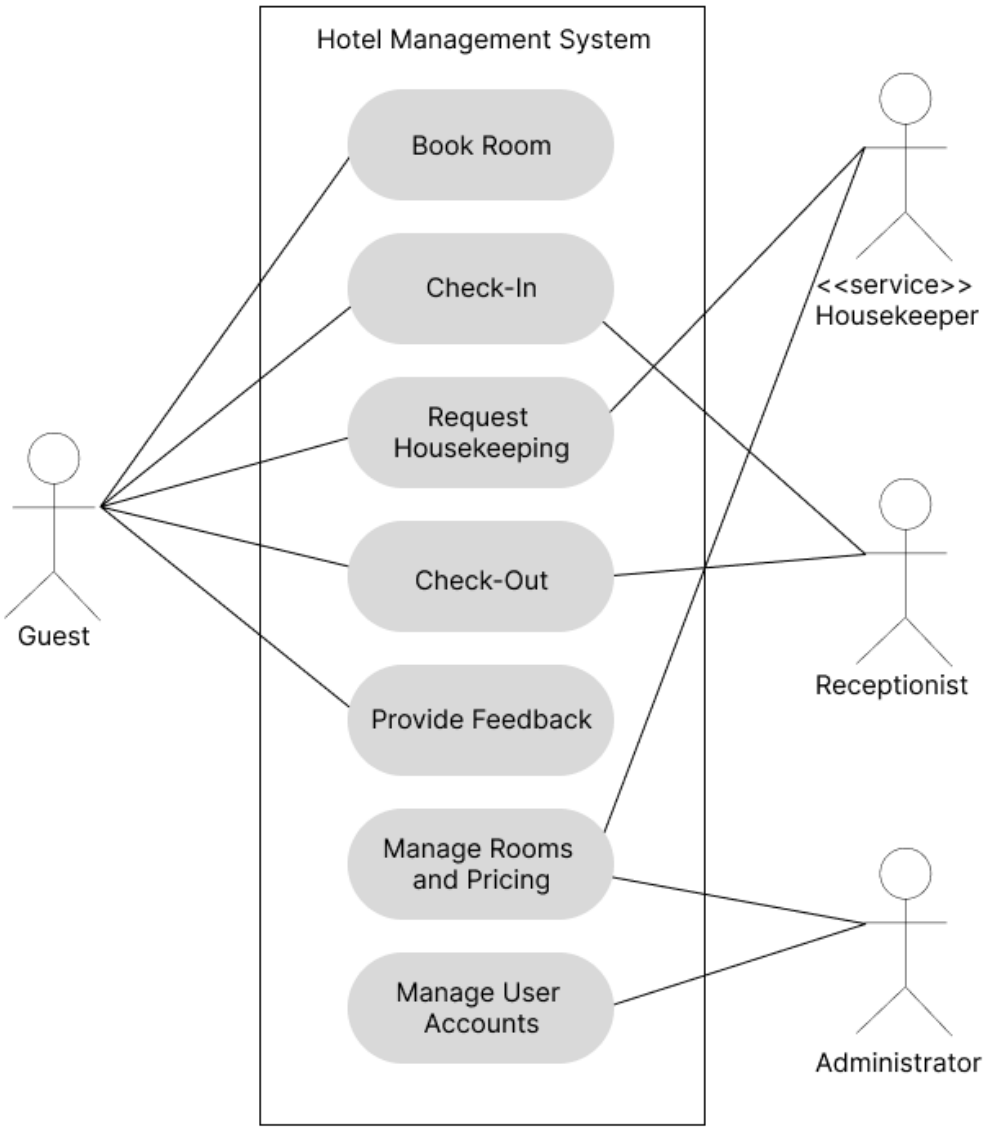
11 SYSTEM DESIGN

11.1 USE CASE DIAGRAMS



11.1.1 ACTORS

Guest: Engages with the facility within the hotel in making a reservation, checking in, checking out, making requests and providing feedback.

Receptionist: Handles guests check in and check out, and deals with room allocations and reservations.

Housekeeper: Provides housekeeping status and accompanies requests made by the department.

Administrator: Controls configurations within the system, the details of the rooms, and accounts created by users.

11.1.2 USE CASES

Book Room: Guests can look and make reservations for rooms that they desire. Receptionists can do manual bookings for walk in customers.

Check-In: Guests are given keys and recorded as checked in by the receptionist.

Check-Out: The receptionist changes the room status after the payment process.

Request Housekeeping: Hotel guests can request such services as cleaning and more amenities.

Update Room Status: Housekeeping staff determine the state of a room as either clean or in need of repair.

Provide Feedback: Guests can write comments or feedback based on the hotel's services.

Manage Rooms and Pricing: The room catalogue, the set prices and promos are controlled by the administrator.

Manage User Accounts: The administrators assign staff roles and who can see what services and secure the system.

11.1.3 FLOW DESCRIPTION

11.1.3.1 GUEST USE CASES

Book Room: Prospective guests look at availability, choose the time, and make the booking.

Request Housekeeping: Requests for cleaning and more materials are made by guests through the system or on telephone by contacting the front desk.

Provide Feedback: Feedback can be provided after checking out or even through the system interface.

11.1.3.2 RECEPTIONIST USE CASES

Check-In: Rooms are allocated to the guests and bookings confirmed and rooms assigned by the receptionists

Check-Out: Several tasks need to be performed, such as reviewing the balance of accounts, taking payments and charging the appropriate status against the room.

11.1.3.3 HOUSEKEEPER USE CASE

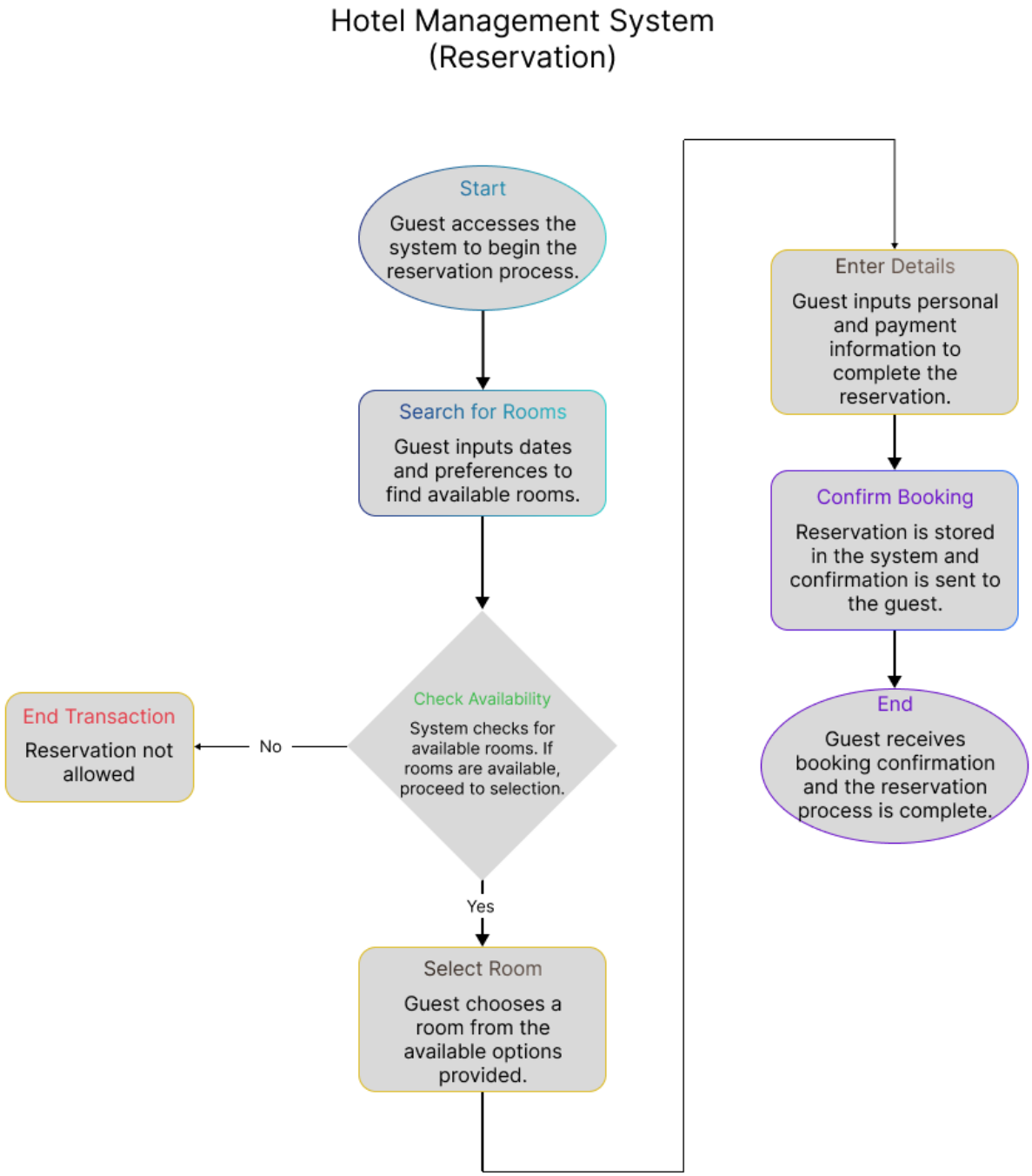
Update Room Status: The room had been cleaned or checked and the Housekeeper was able to input its status onto the system.

11.1.3.4 ADMINISTRATOR USE CASE

Manage Rooms and Pricing: The administrator includes details of newly available rooms, revises the rates and puts special offers into effect.

Manage User Accounts: The administrator can include or delete staff accounts and their roles as well.

11.2 ACTIVITY DIAGRAMS



11.2.1 THE PROCEDURE OF MAKING A RESERVATION

11.2.1.1 FLOW DESCRIPTION

Start: The system is accessed by the guest.

Look for available Rooms: The guest gives specific details such as the dates and preferences for a certain room.

Checking the Availability of a Room: The system provides an answer about the rooms available.

Proceed with the process if there are any rooms that are available.

End the process if there are no rooms available.

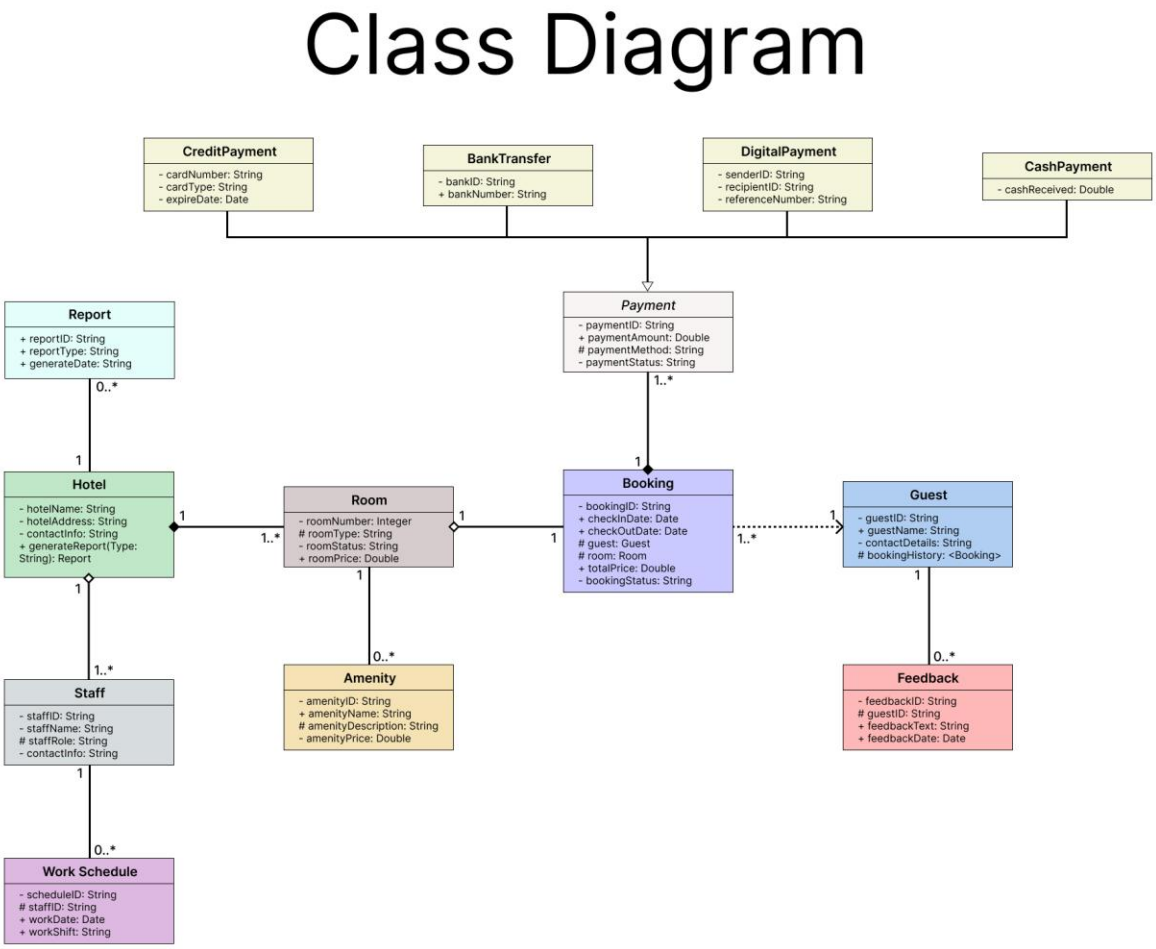
Choose a Room: The room is selected by the guest.

Input the Information: The guest fills personal and billing details.

Making the Reservation: The system logs and reserves the details and sends a message with confirmation of booking.

End: The guest gets a message with confirmation of the initiated booking ending the process.

11.3 CLASS DIAGRAMS



11.3.1 CLASSES, ATTRIBUTES AND RELATIONSHIPS

11.3.1.1 HOTEL

Attributes: hotelName, hotelAddress, contactInfo, generateReport

Relationship: Association with Report class, Aggregation with Staff class, and Composition with Room class.

11.3.1.2 ROOM

Attributes: roomNumber, roomType, roomStatus, roomPrice

Relationship: Composition with the Hotel class, Association with the Amenity class, and Aggregation with the Booking class.

11.3.1.3 STAFF

Attributes: staffID, staffName, staffRole, contactInfo

Relationship: Aggregation with the Hotel class and Association with the Work Schedule class

11.3.1.4 BOOKING

Attributes: bookingID, checkInDate, checkOutDate, #guest, #room, totalPrice, bookingStatus

Relationship: Dependent to the Guest class, Aggregation to the Room class and Composition with the Payment class.

11.3.1.5 FEEDBACK

Attributes: feedbackID, guestID, feedbackText, feedbackDate

Relationship: Association with the Guest class.

11.3.1.6 GUEST

Attributes: guestID, guestName, contactDetails, bookingHistory

Relationship: The Booking class Depends on the guest class that triggers the system to start.

11.3.1.7 AMENITY

Attributes: amenityID, amenityName, amenityDescription, amenityPrice

Relationship: Association with the Room class.

11.3.1.8 REPORT

Attributes: reportID, reportType, generateDate

Relationship: Association with the Hotel class

11.3.1.9 PAYMENT

Attributes: paymentID, paymentAmount, paymentMethod

Relationship: Has Inheritance relationship with CreditPayment class, BankTransfer class, DigitalPayment class, and CashPayment class.

11.3.1.10 WORK SCHEDULE

Attributes: scheduleID, staffID, workDate, workShift

Relationship: Association with the Staff class

11.3.2 SYSTEM FLOW

11.3.2.1 GUEST INITIATES BOOKING

The guest fills up the provided form about their details (for instance, guestName, contactDetails) before they can proceed to select a room.

11.3.2.2 ROOM ASSIGNMENT

After completing the form, the guest can now choose specific room that is currently available.

11.3.2.3 BOOKING CREATION

For the guest, a booking is made by connecting a room that has been picked out by them. This incorporates the (checkInDate, checkOutDate, and bookingStatus) among others.

11.3.2.4 PAYMENT PROCESSING

A guest is required to make a payment after confirmation of booking. There are many ways the guest can settle their payment like:

- Credit Payment
- Bank Transfer
- Portrait Settlements
- Cash Payment

11.3.2.5 AMENITY SELECTION (IF AVAILABLE)

During booking or after the booking is done, the guest may opt for some additional amenities like spa treatment or eating at the hotel which are charged separately. The amenity cost is provided as an addition to the total price.

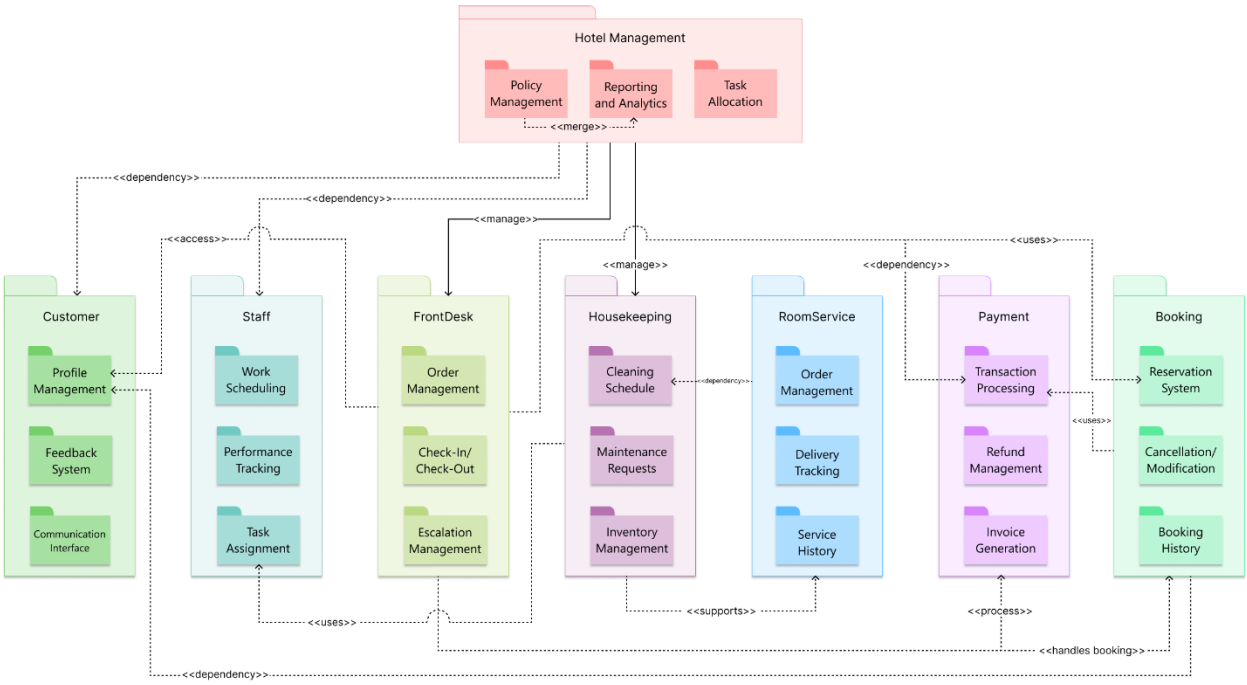
11.3.2.6 STAY AT HOTEL

The guest arrives for check in on checkInDate provided and then checks out of the same specific room on checkOutDate.

Feedback Submission (optional):

Upon check out, the guest can leave a review about their experience in the hotel.

11.4 PACKAGE DIAGRAM



11.4.1 PACKAGE ORGANIZATION

11.4.1.1 HOTEL MANAGEMENT

Policy Management: Establishes principles and service rules.

Reporting and Analytics: Contributes to the performance measurement (e.g. staff efficiency, room occupancy level).

Task Allocation: Speaks about authority delegation to Staff, FrontDesk, etc.

11.4.1.2 STAFF

Work Scheduling: Provides shift management system for staff.

Performance Tracking: Monitors the effectiveness of staff and measure the clients' opinions.

Task Assignment: Daily assignments including – cleaning assignments, requests for services (room service etc.).

11.4.1.3 CUSTOMER

Profile Management: Records information about the customers, their interests and experience with the company.

Feedback System: Captures rating and comments about the provided services from the clients.

Communication Interface: Enables interaction with the FrontDesk such as requests, inquiries, etc.

11.4.1.4 FRONT DESK

Check-In/Check-Out: Concerned with the reception of visitors and their departure.

Inquiries Handling: Supporting customers' questions and requests.

Escalation Management: Involves issues that can only be solved with the intervention of senior authority or Hotel management.

11.4.1.5 BOOKING

Reservation System: Capable of room bookings and checking room availability.

Cancellation/Modification: Designed to conduct any changes in the bookings done by the clients.

Booking History: Remembers what the past booking was about.

11.4.1.6 PAYMENT

Transaction Processing: Payments are made by credit card, online mode or cash.

Refund Management: Handles refund payments based on cancellations/interruption of subscription.

Invoice Generation: Makes invoices and issue them to customers.

11.4.1.7 ROOM SERVICE

Order Management: Manage order requests from a guest regarding the menu, beverages and any other facility.

Delivery Tracking: Keeps track of in-room deliveries and their status.

Service History: Keeps a record of services/ request that have already been done.

11.4.1.8 HOUSEKEEPING

Cleaning Schedule: Organizes and allocates daily cleaning duties to both occupied and vacant rooms.

Maintenance Requests: Deals with the service and repairs and maintenance of the room.

Inventory Management: Monitors availability of the stock and various room supplies.

11.4.2 TOP-LEVEL PACKAGE: HOTEL MANAGEMENT

This section has procedures outlines for all levels; delegation of responsibilities, reporting and, analytics.

11.4.3 SUB-PACKAGES

Staff: Assign in relation to scheduling, task distribution and subordinate performance.

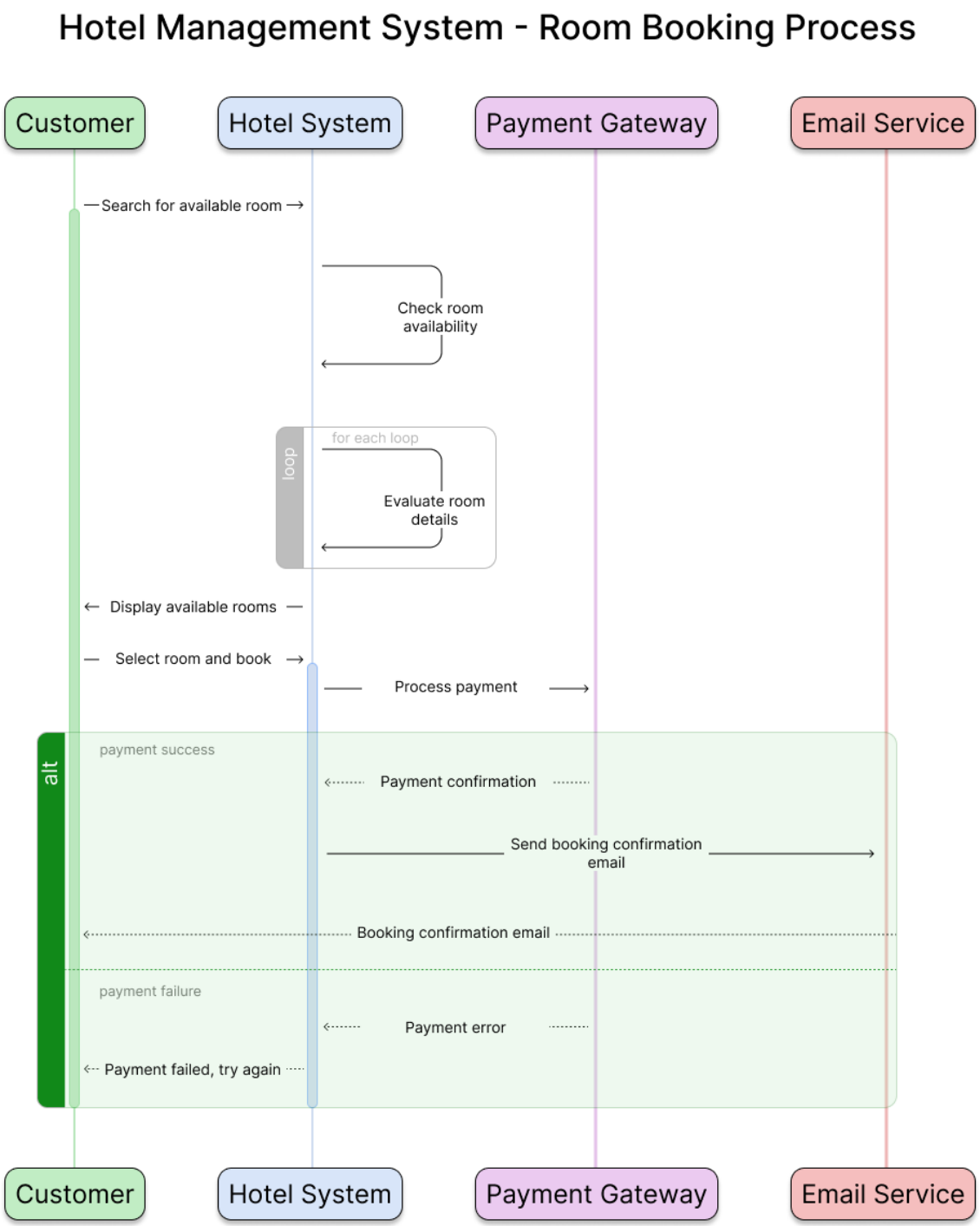
FrontDesk: Interact with the customer during the registration, clients' queries processes etc.

Customer: Manage the customer profile and feedback.

RoomService and Housekeeping: In charge for preparation of the room for room service and its delivery.

Booking and Payment: Deals with booking activities and the payment activities.

11.5 SEQUENCE DIAGRAMS



11.5.1 ROOM BOOKING PROCESS

11.5.1.1 CUSTOMER INTERACTION

- The process is initiated when the customer wishes to locate an empty room.
- Once the room has been confirmed as available, its description is provided to the customer.
- The customer decides upon a room and begins the booking process.

11.5.1.2 HOTEL SYSTEM OPERATIONS

- When there are lots of free rooms, all of them are enhanced by detailed room descriptions stored in the system and these are displayed to the customers.
- If a room has already been selected, the preferred occupation is booked, and payment is made.

11.5.1.3 PAYMENT PROCESS

The request for payment is generated by the system for the Payment Gateway.

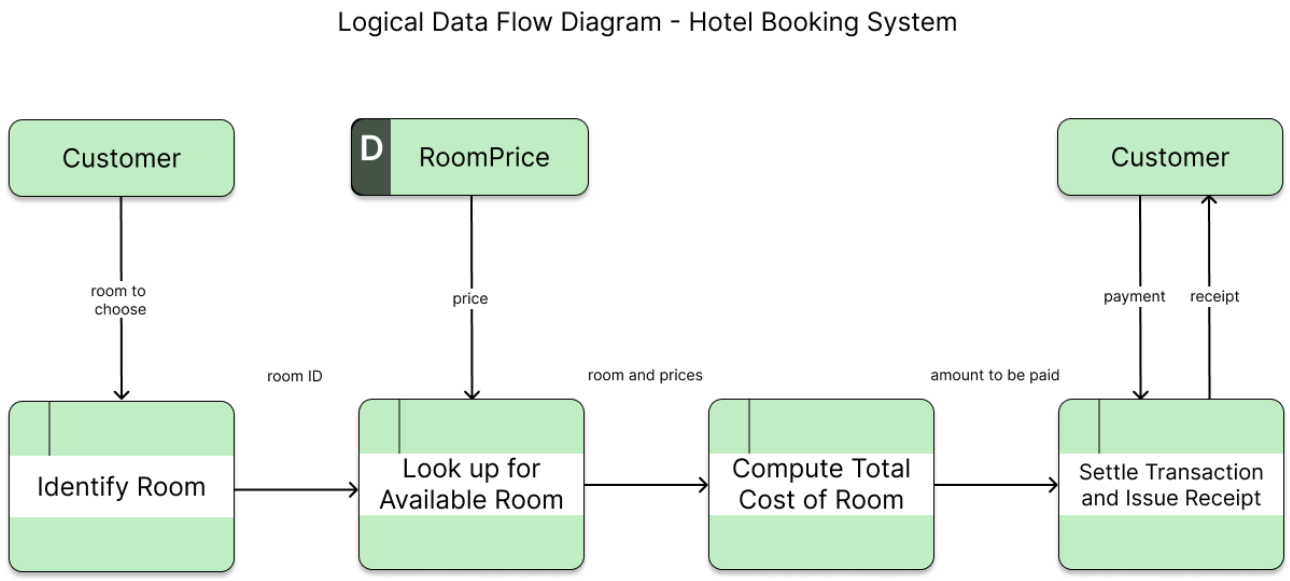
The possible possibilities are:

- Payment Processed: The Payment Gateway processes the payment, and the Hotel System automatic booking confirmation email has been sent via Email service. The email then reaches the customer's inbox finalizing the process.
- Payment Not Processed: In case the payment has not gone through, the customer would be highlighted by the system which would drive them to attempt for the second time.

11.5.1.4 EMAIL SERVICE

When the booking has been made and the payment has been completed, the Email Service will respond to the customer with an email confirming their reservation.

11.6 DATA FLOW DIAGRAMS (DFD)



11.6.1 DIAGRAM FLOW

11.6.1.1 IDENTIFY ROOM

The first step in the procedure is where the user selects a room which they would like to book.
The room ID selected is picked by the system for use.

11.6.1.2 LOOK UP FOR AVAILABLE ROOM

To find the room chosen by the user and its cost, the system goes to the RoomPrice data base.
The searches of the room and price availabilities are done, and results are moved to the next step.

11.6.1.3 COMPUTE TOTAL COST OF ROOM

The system can determine the total amount by the price of the room.
Determines and gets ready the total cost for payment.

11.6.1.4 SETTLE TRANSACTION AND ISSUE RECEIPT

The last step is completed: the client paid the full amount for all transactions.
The receipt is provided to the customer immediately after the payment has occurred.

11.6.2 KEY ELEMENTS IN THE DIAGRAM

11.6.2.1 ENTITIES

Customer: This is the outside actor which triggers the system.

11.6.2.2 PROCESSES

These are actions taken in the system. To include Identify Room, Look up for Available Room, Compute Total Cost of Room, Settle Transaction and issue receipt.

11.6.2.3 DATA STORE

RoomPrice: Contains the information pertaining to the rooms and their prices.

11.6.2.4 DATA FLOW

The arrows show what is moving in between entities, processes and datastores. For instance:

room to choose moves from a customer to an Identify Room process.

''Price'' is acquired from the RoomPrice database.